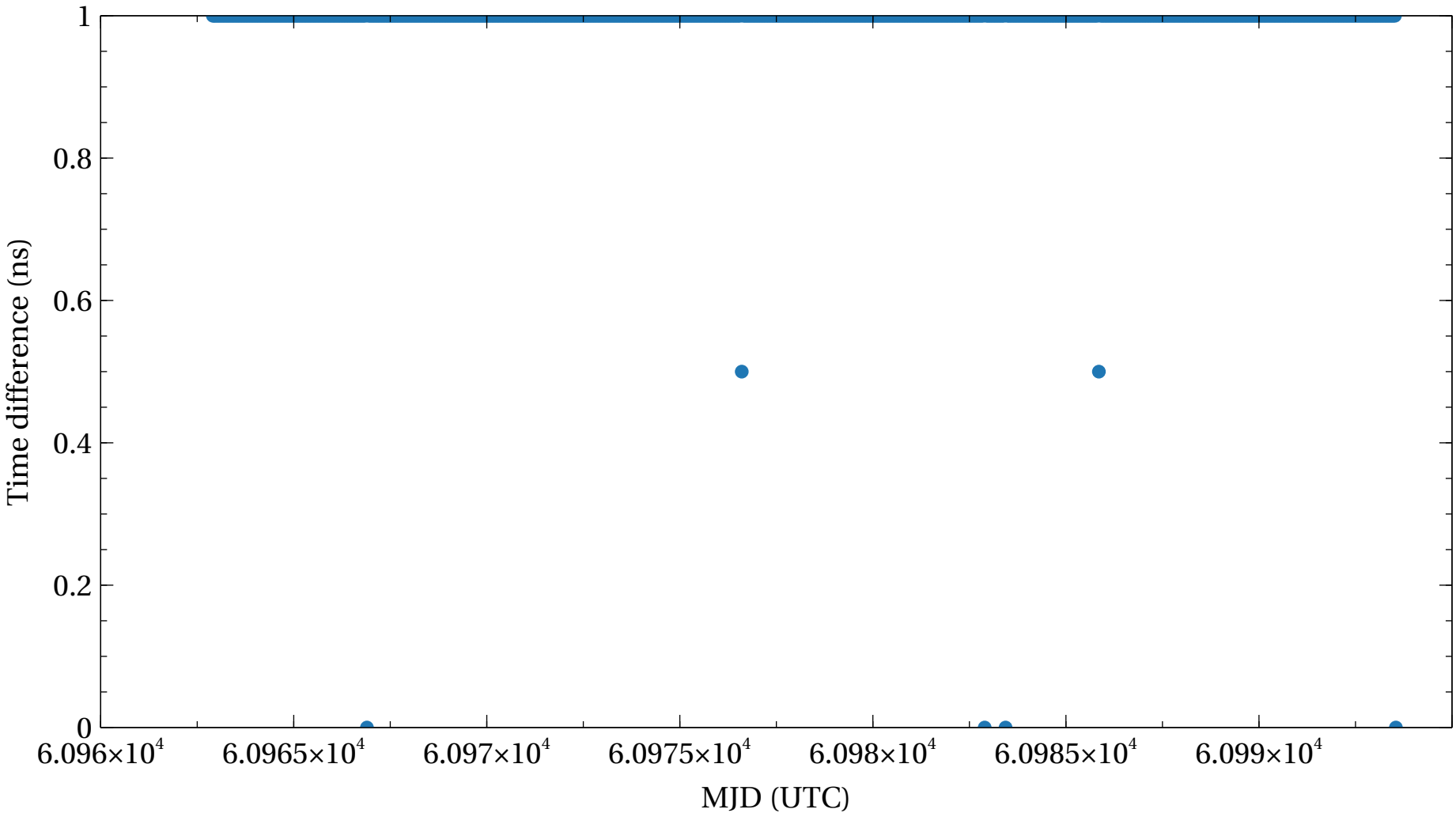
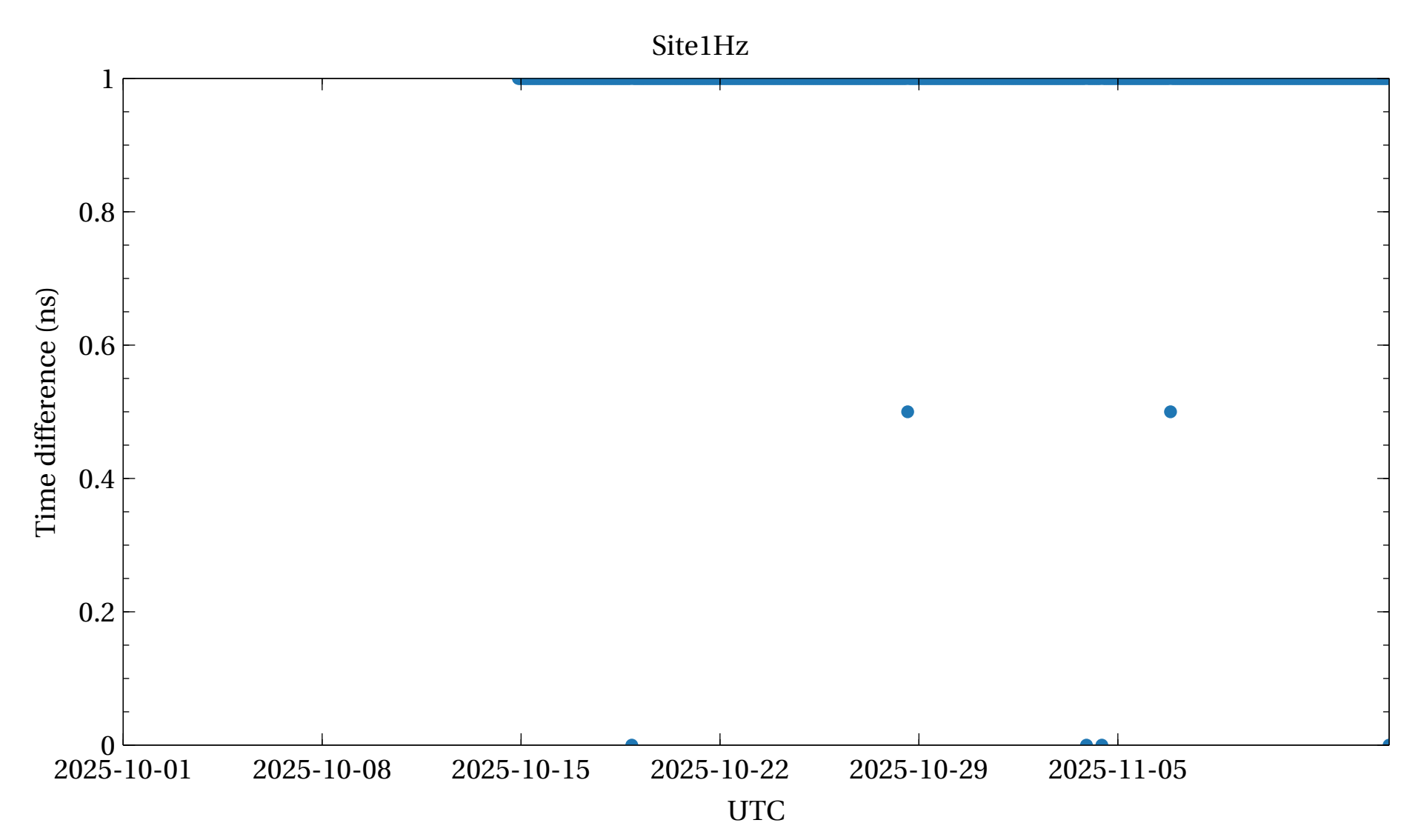
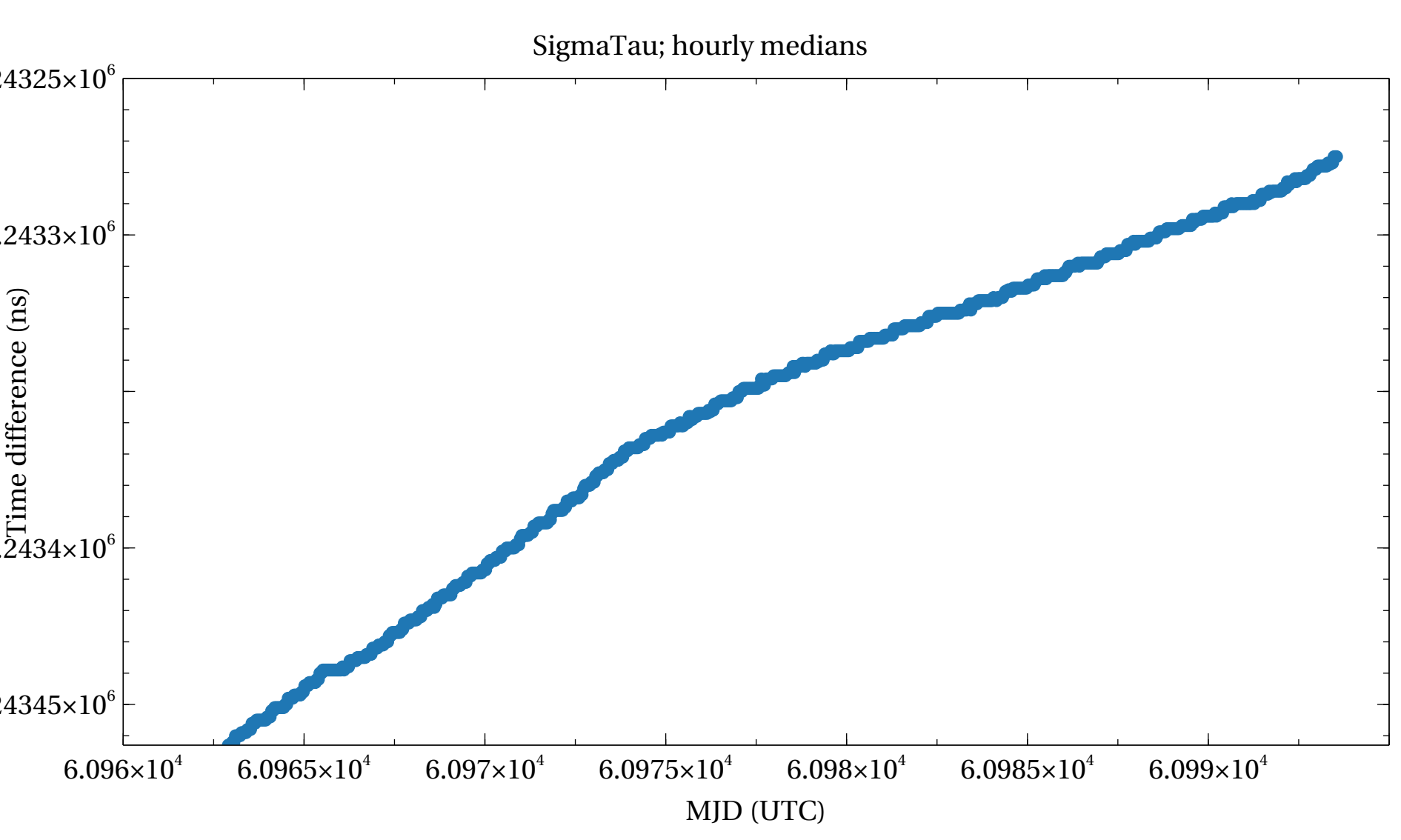
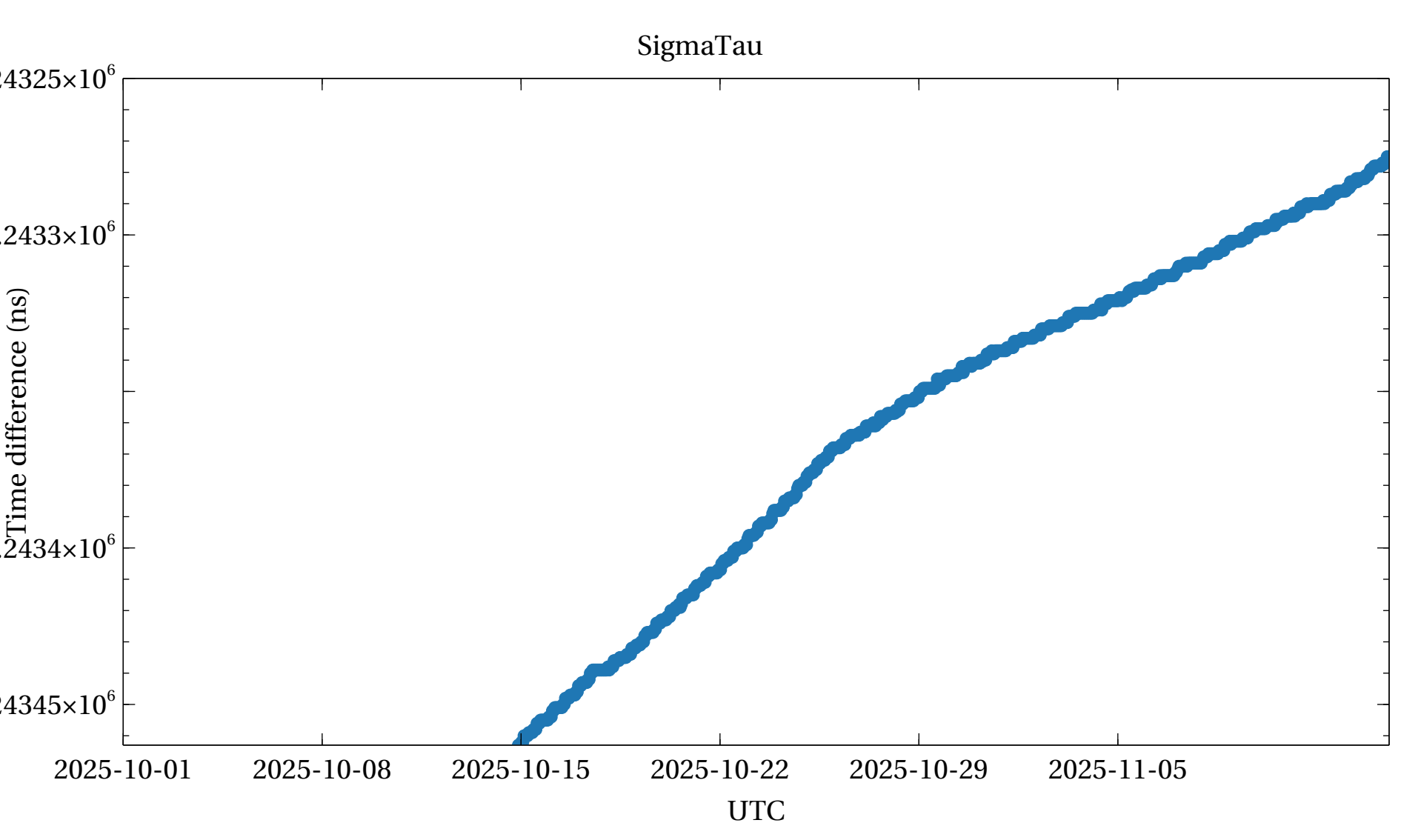
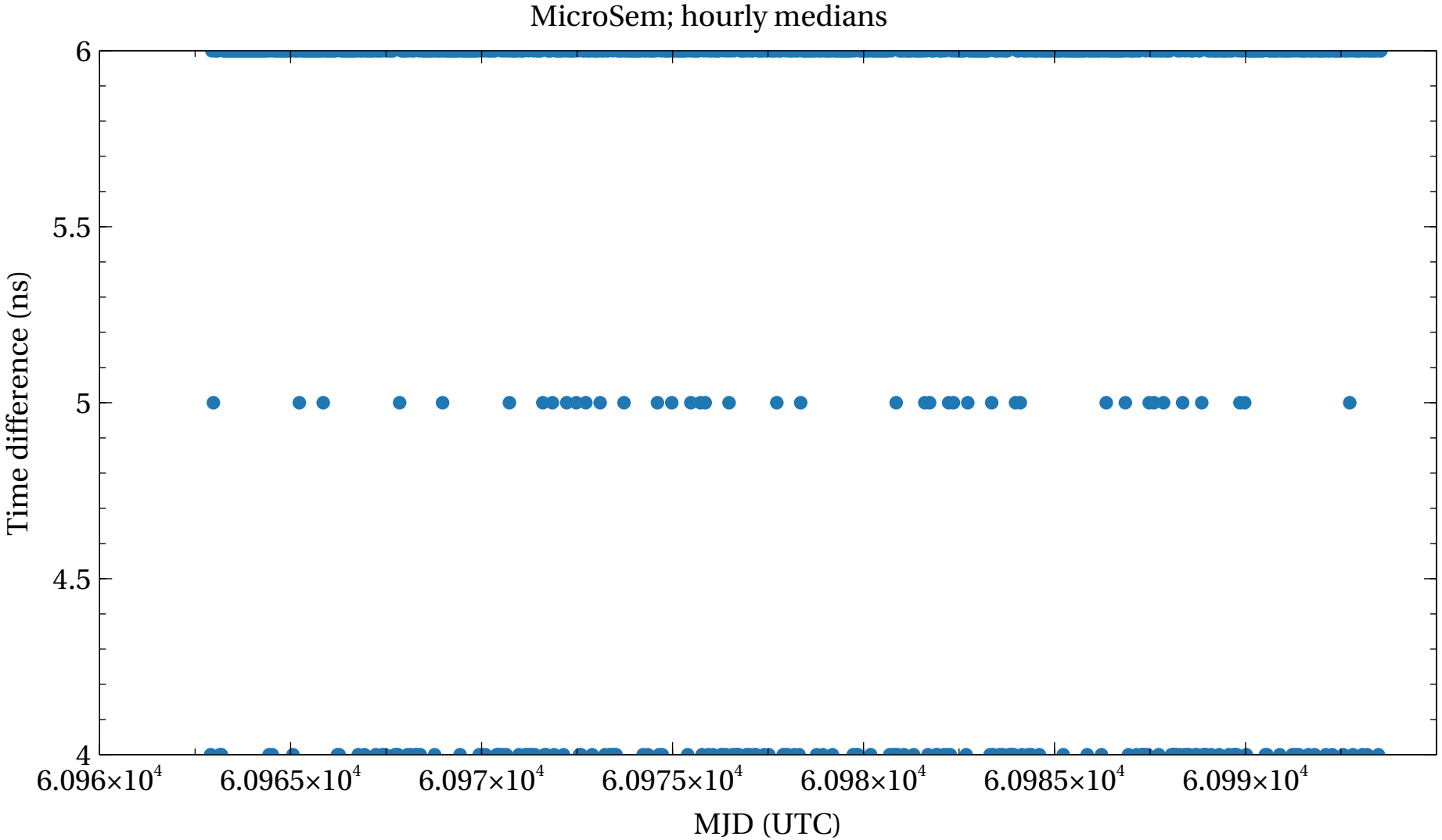
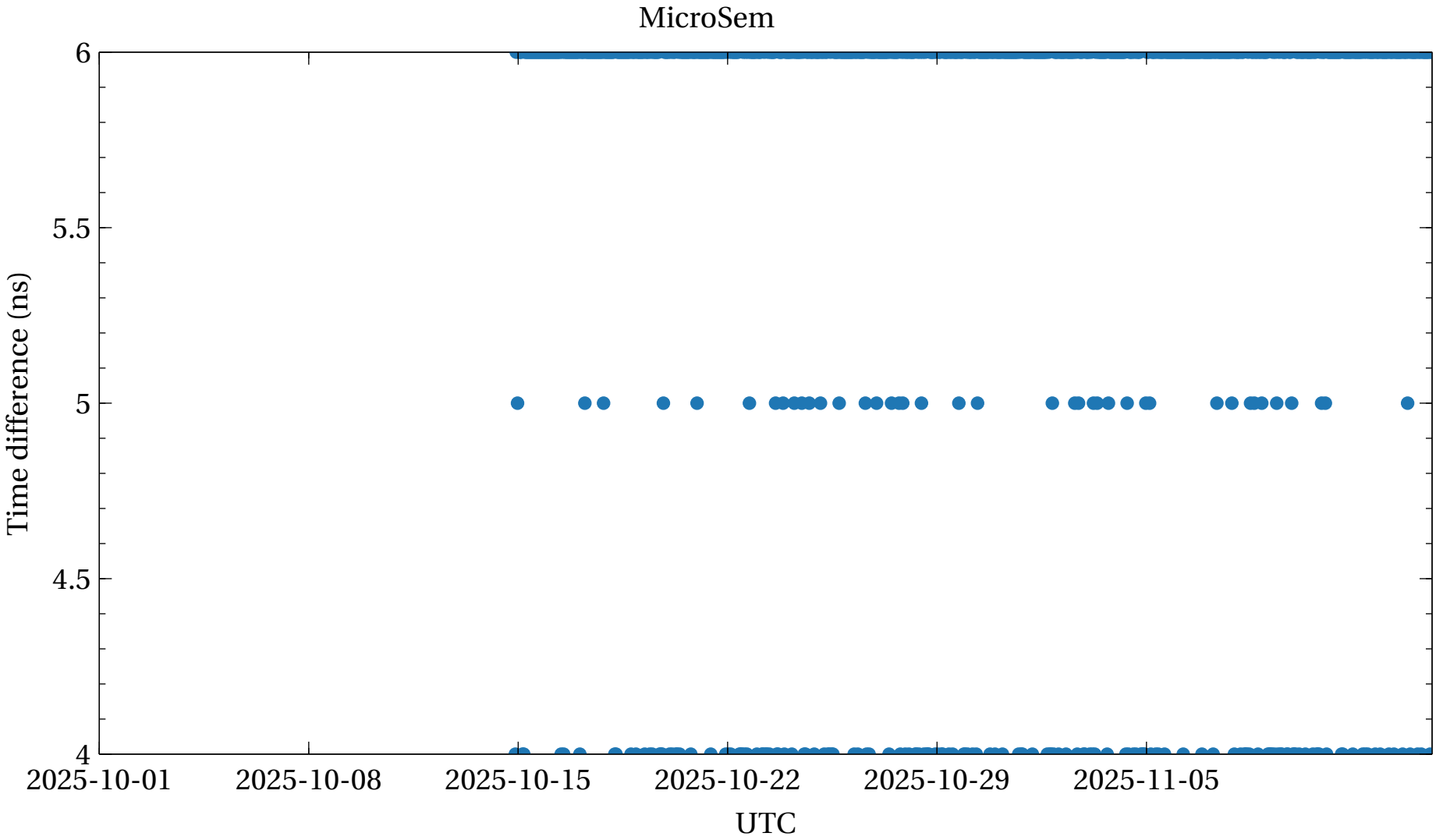

Site1Hz; hourly medians
Time difference (ns)
1
0.8
0.6
0.4
0.2
0
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)

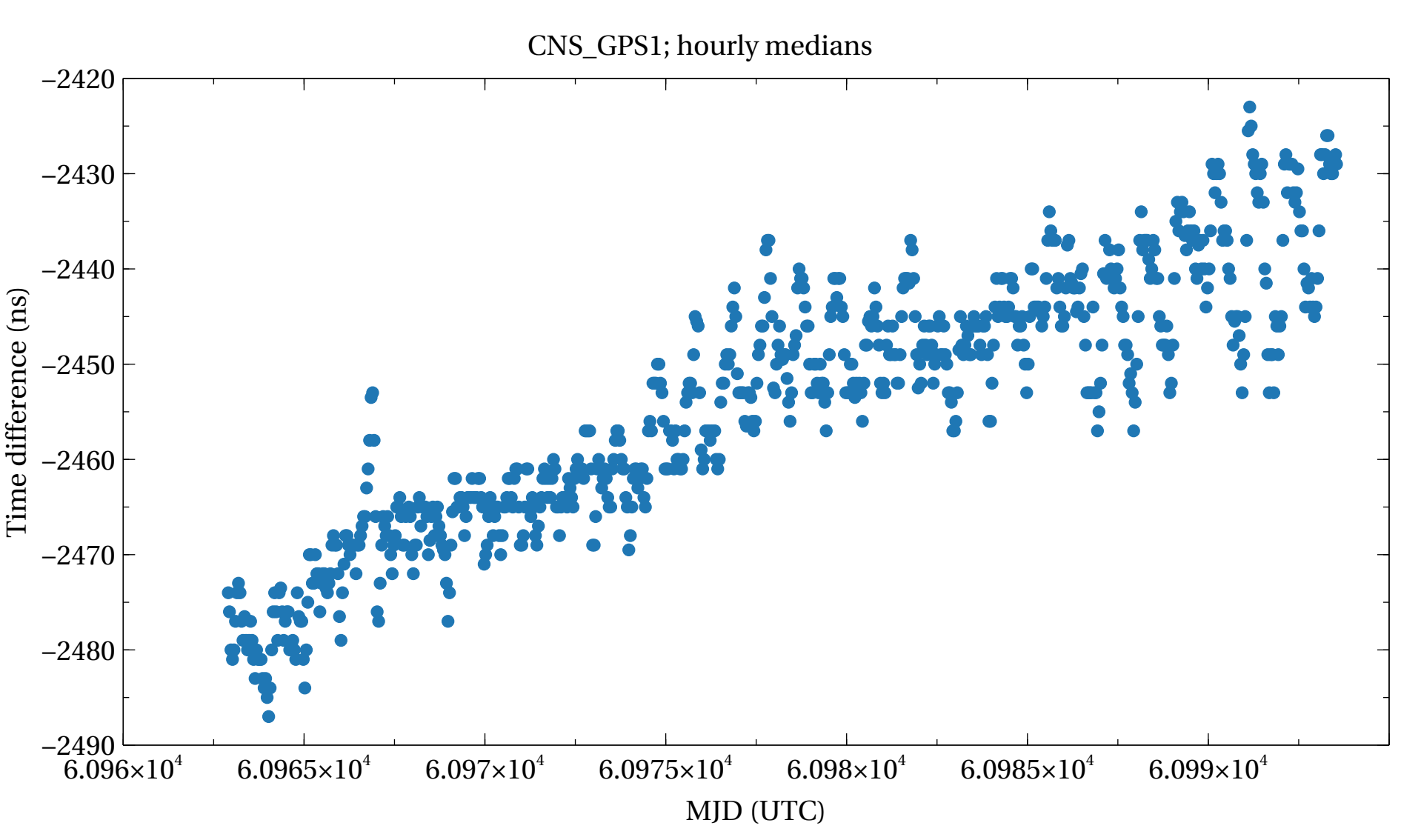

Site1Hz
Time difference (ns)
1
0.8
0.6
0.4
0.2
0
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC

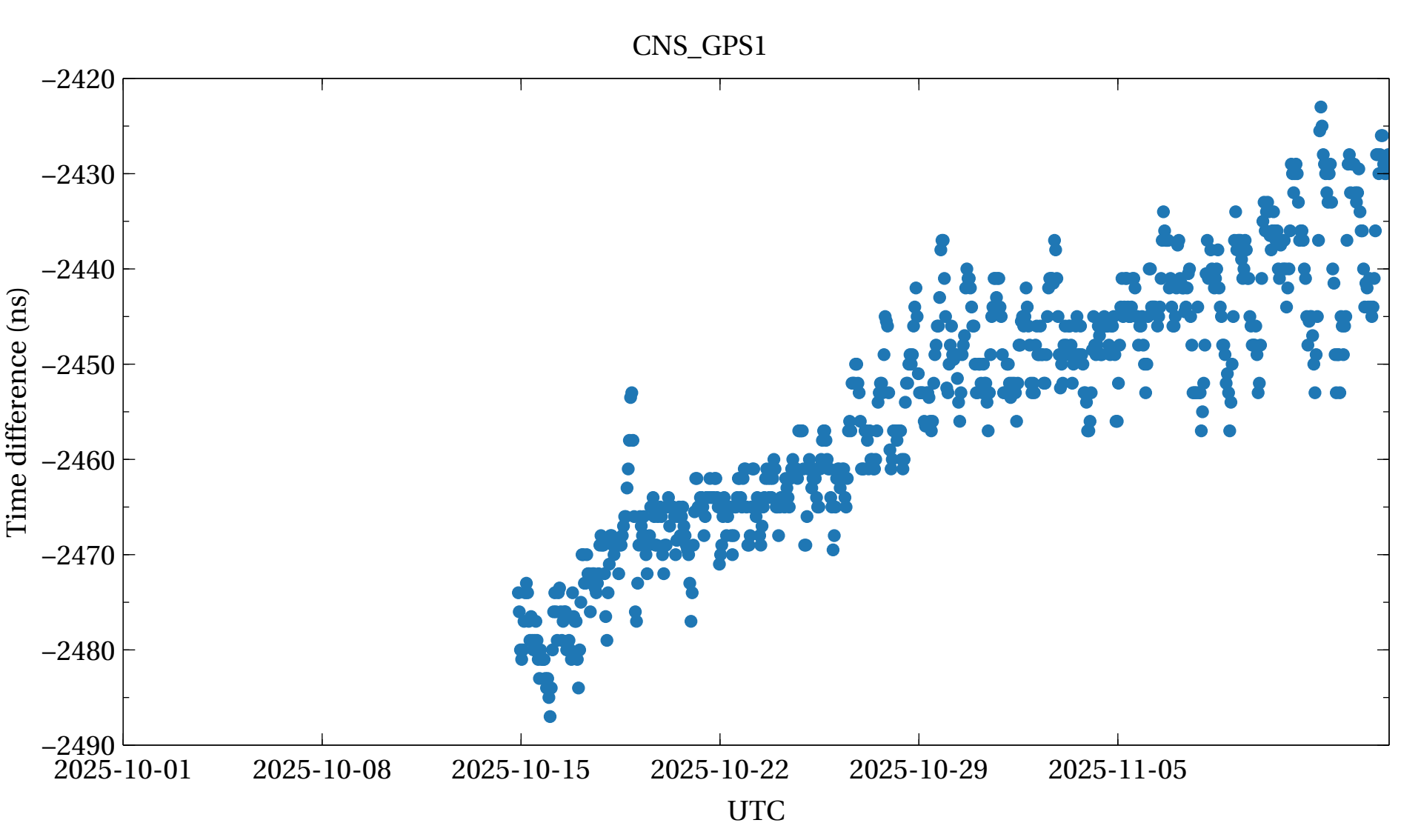

SigmaTau; hourly medians
4325×10^6
2433×10^6
2434×10^6
4345×10^6
Time difference (ns)
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)

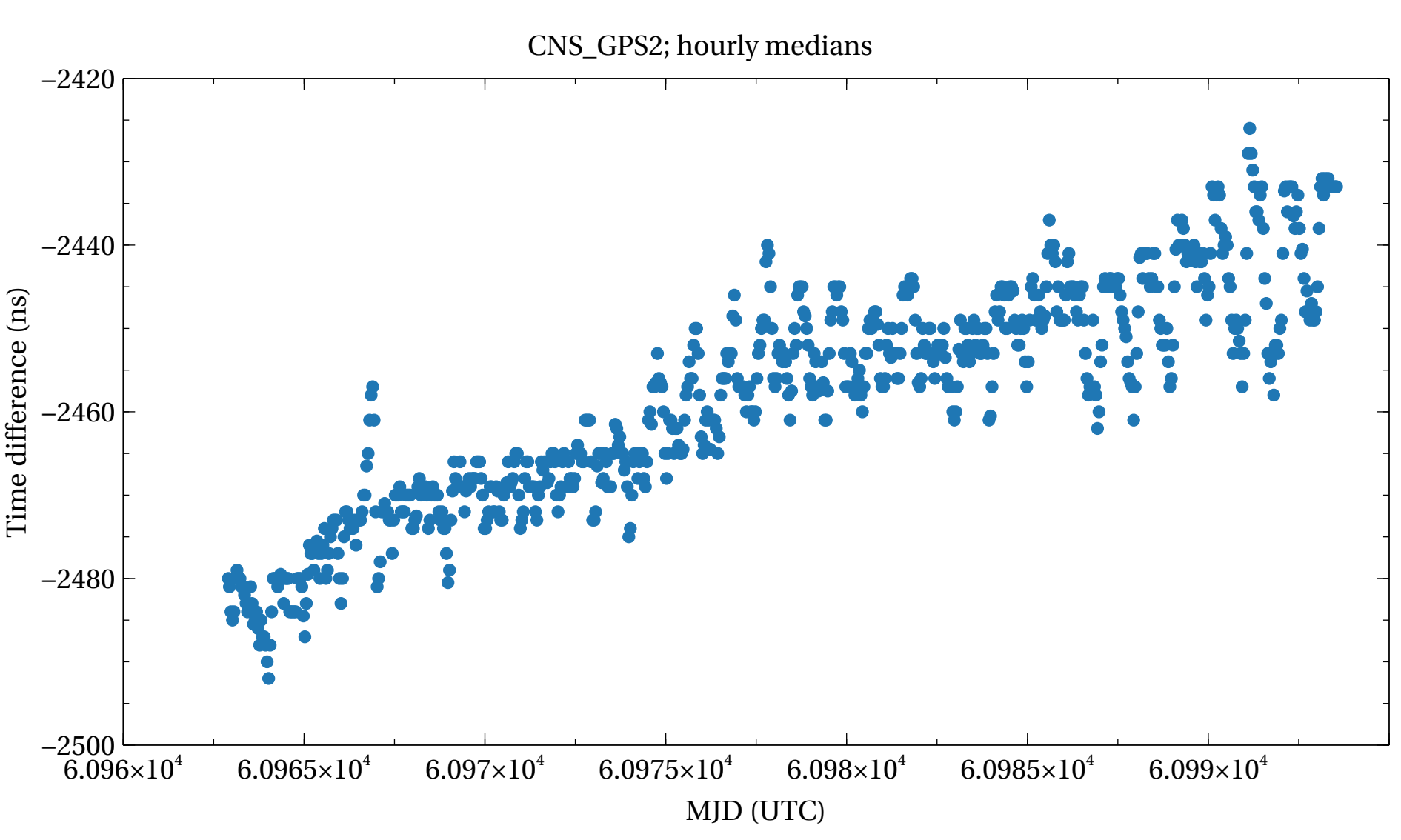

SigmaTau
4325×10^6
2433×10^6
2434×10^6
4345×10^6
Time difference (ns)
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC

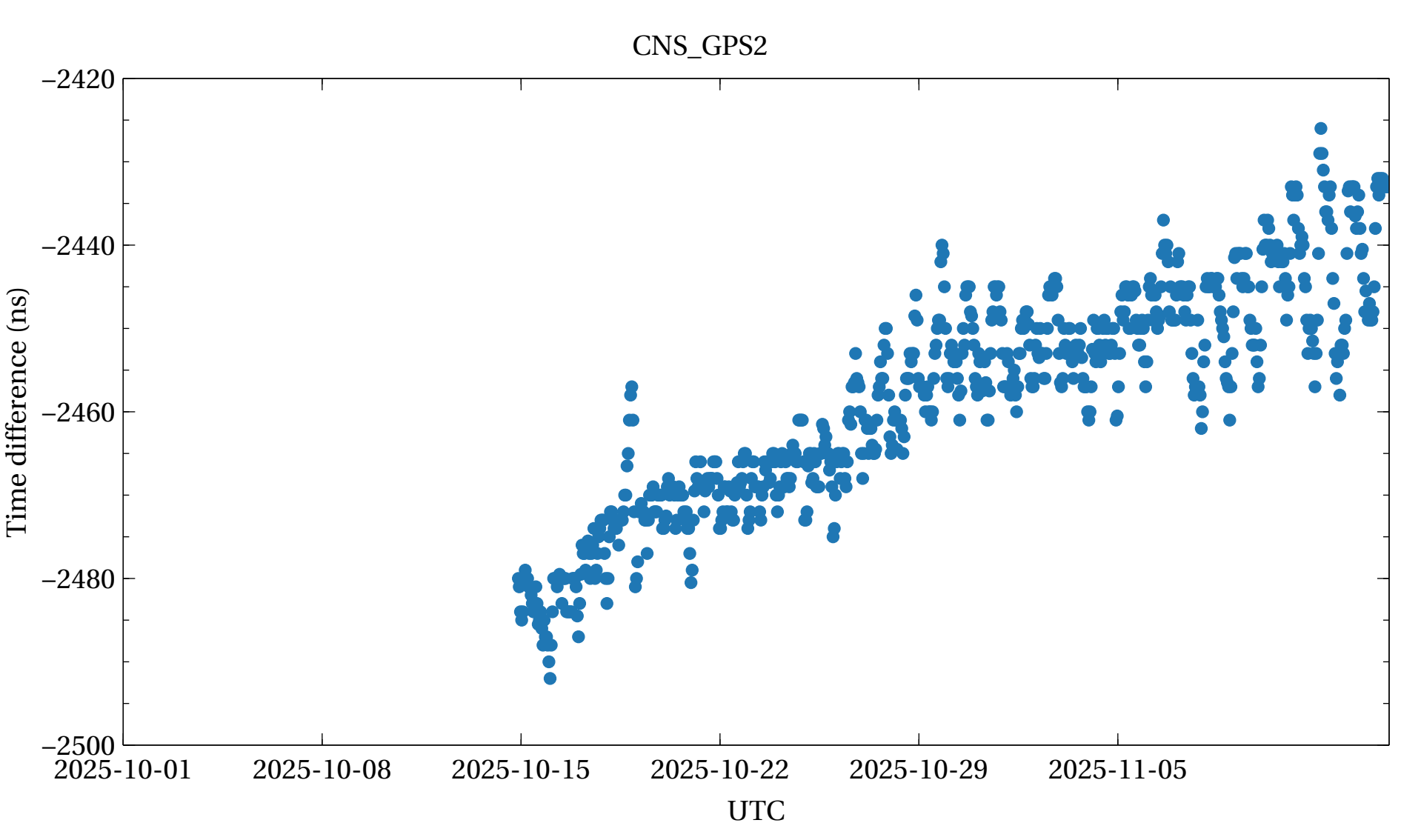

MicroSem; hourly medians
Time difference (ns)
6
5.5
5
4.5
4
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)

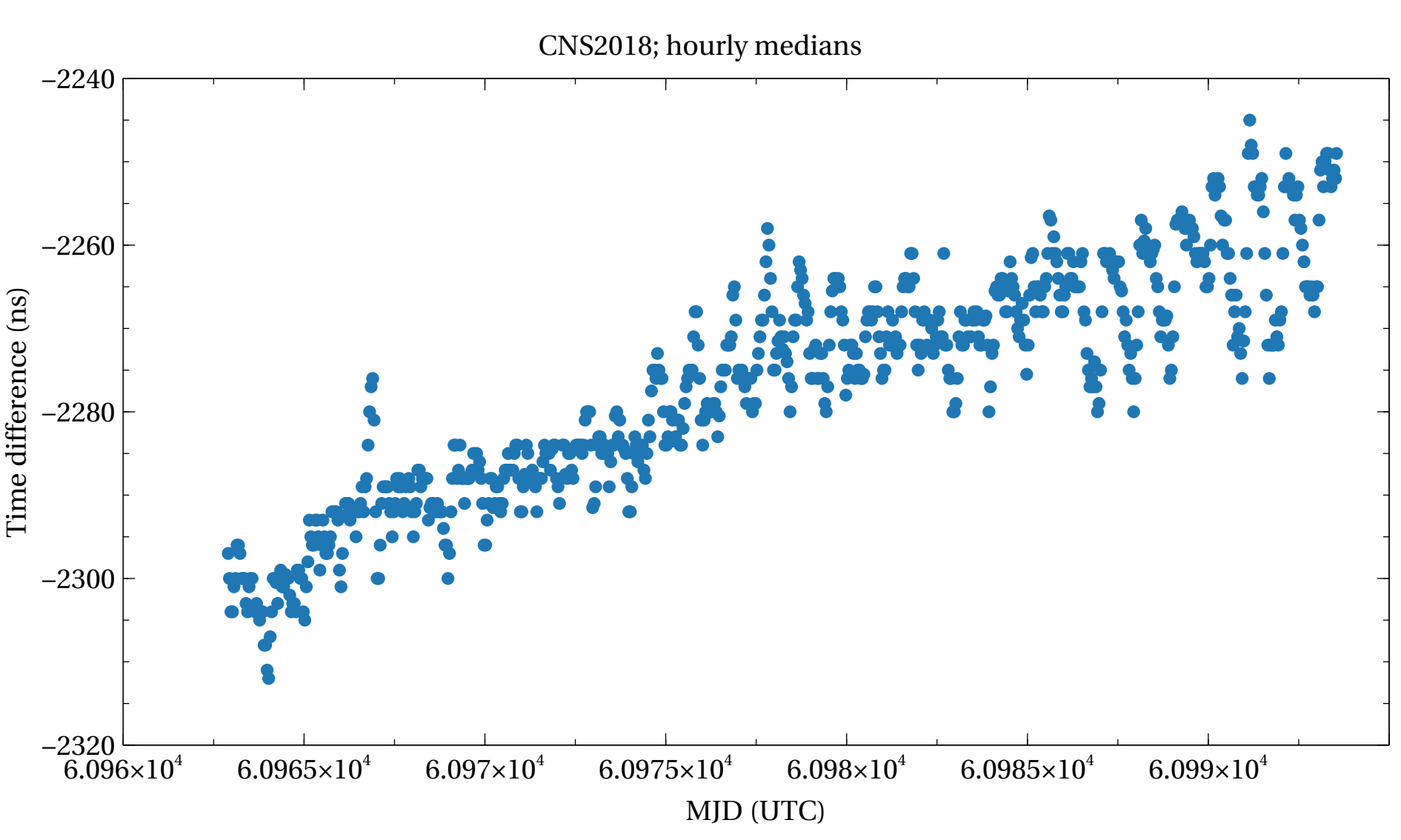

MicroSem
6
5.5
5
4.5
4
Time difference (ns)
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC

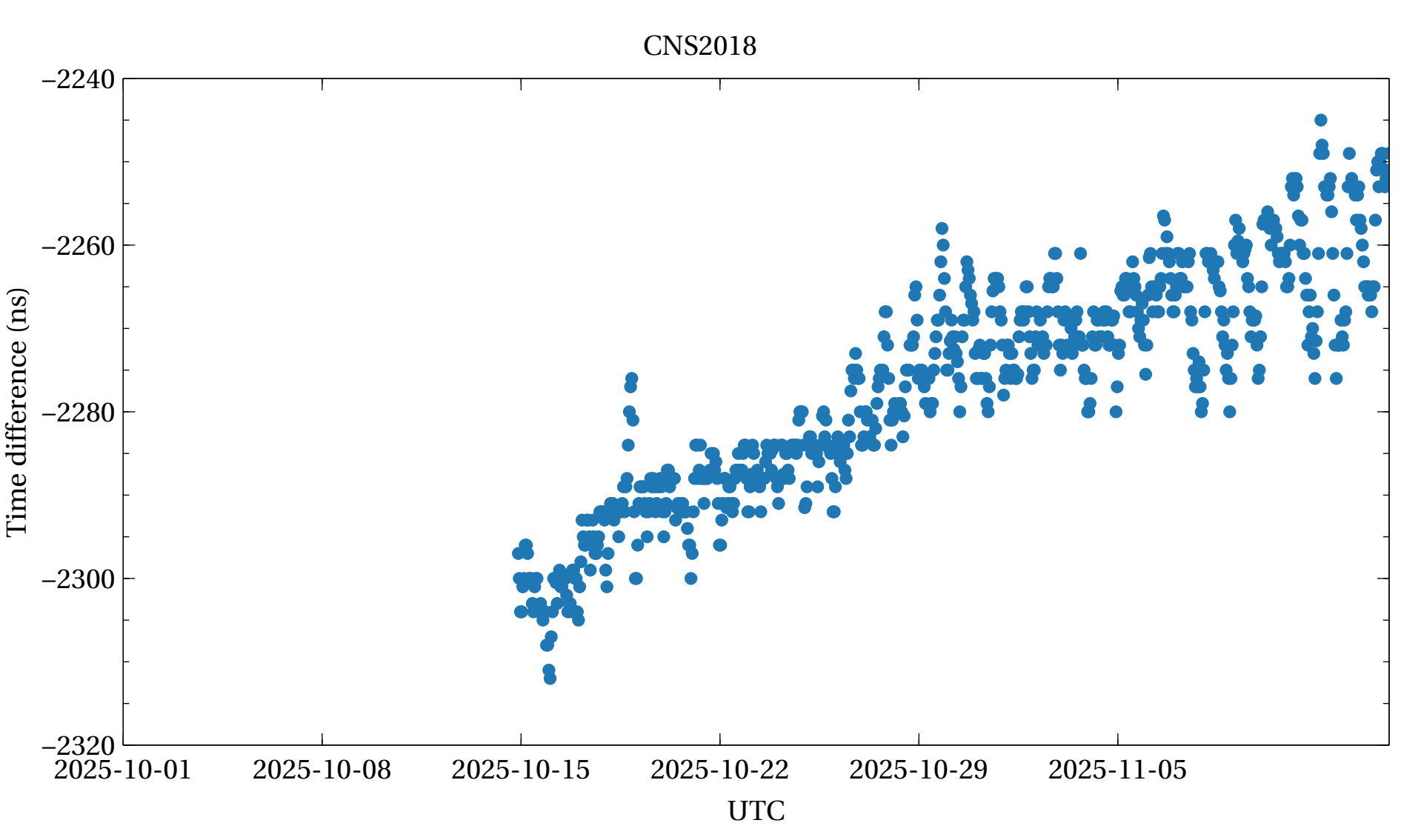

CNS_GPS1; hourly medians
−2420
−2430
−2440
−2450
−2460
−2470
−2480
−2490
Time difference (ns)
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)

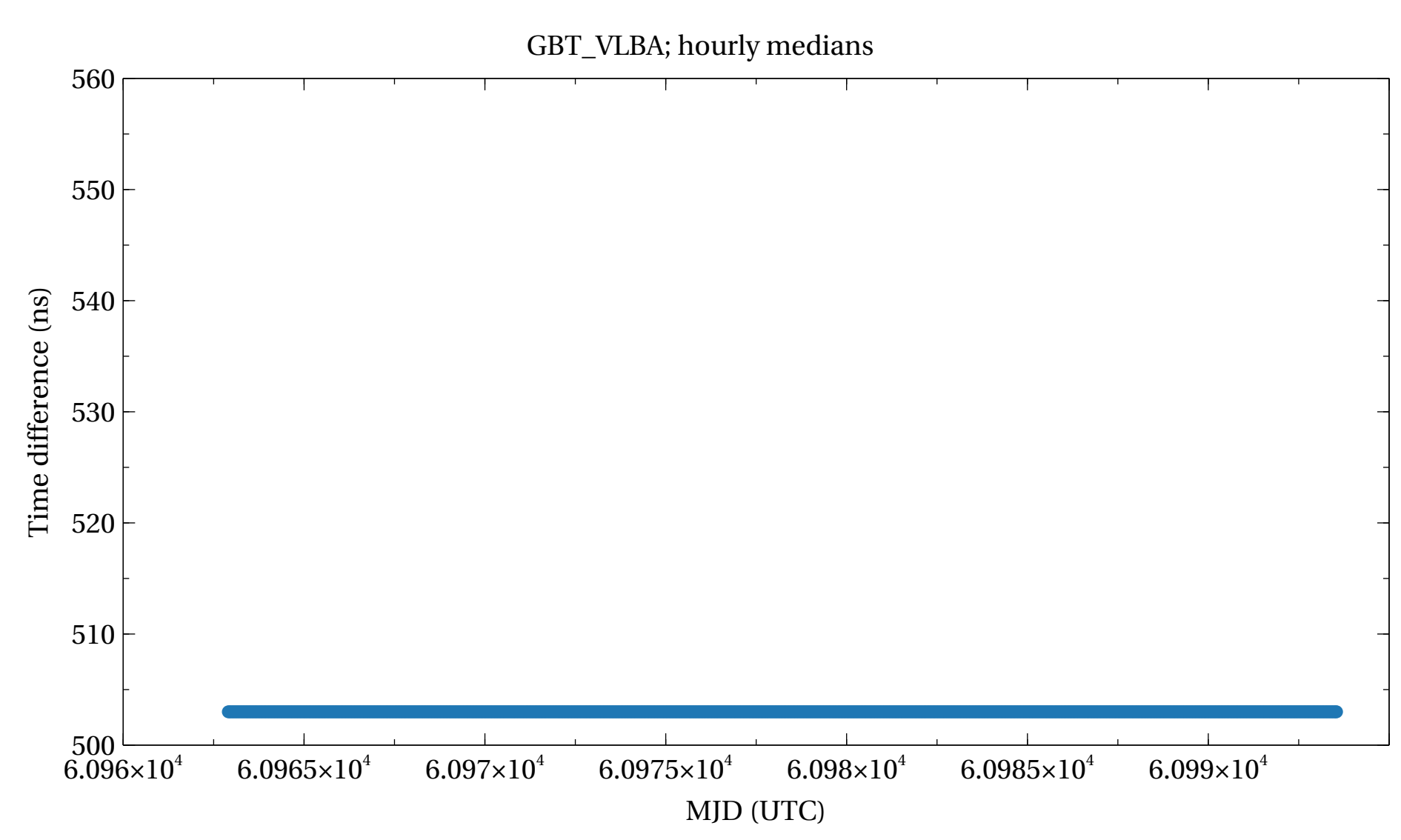

CNS_GPS1
−2420
−2430
−2440
−2450
−2460
−2470
−2480
−2490
Time difference (ns)
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC

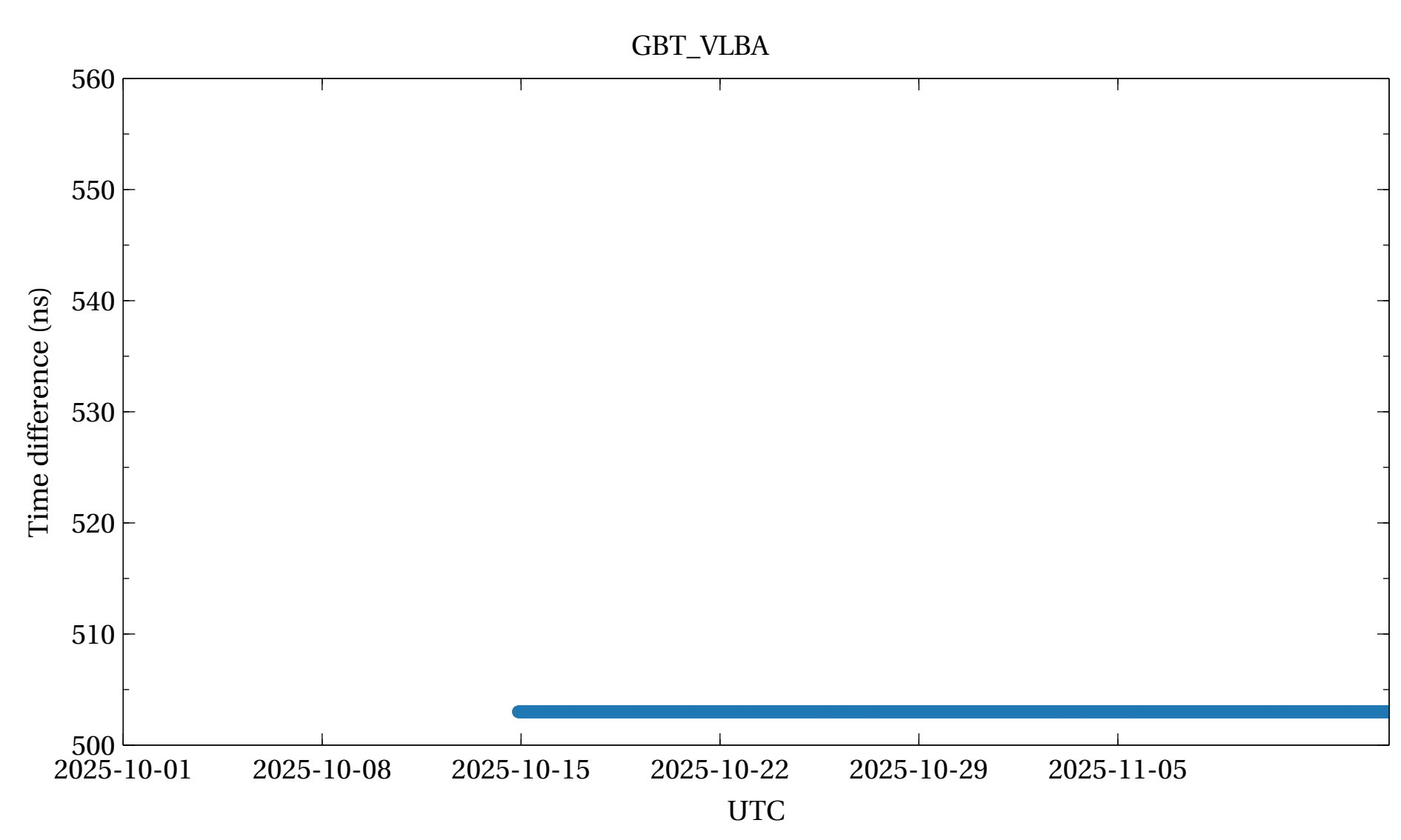

CNS_GPS2; hourly medians
−2420
−2440
−2460
−2480
−2500
Time difference (ns)
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)

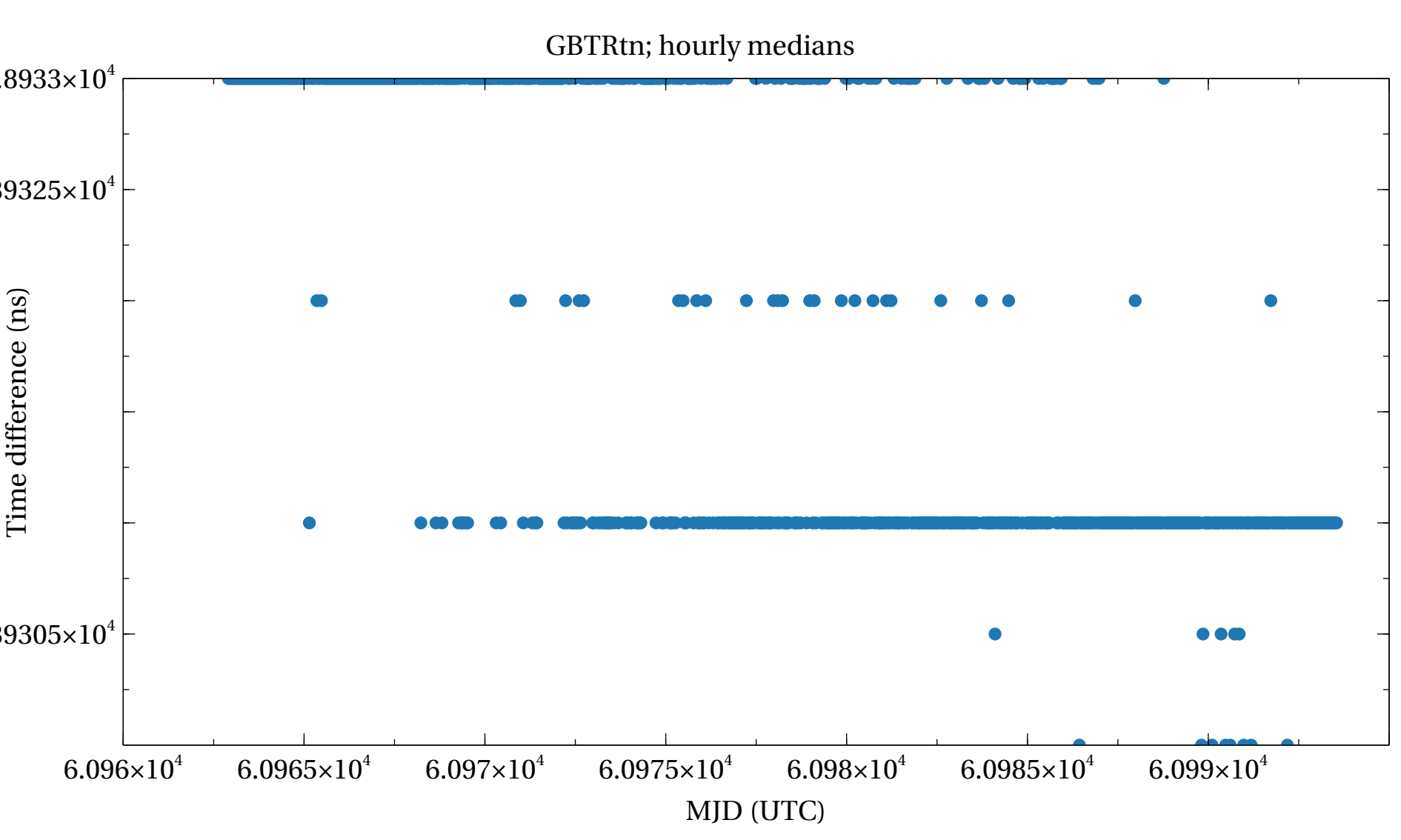

CNS_GPS2
−2420
−2440
−2460
−2480
−2500
Time difference (ns)
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC

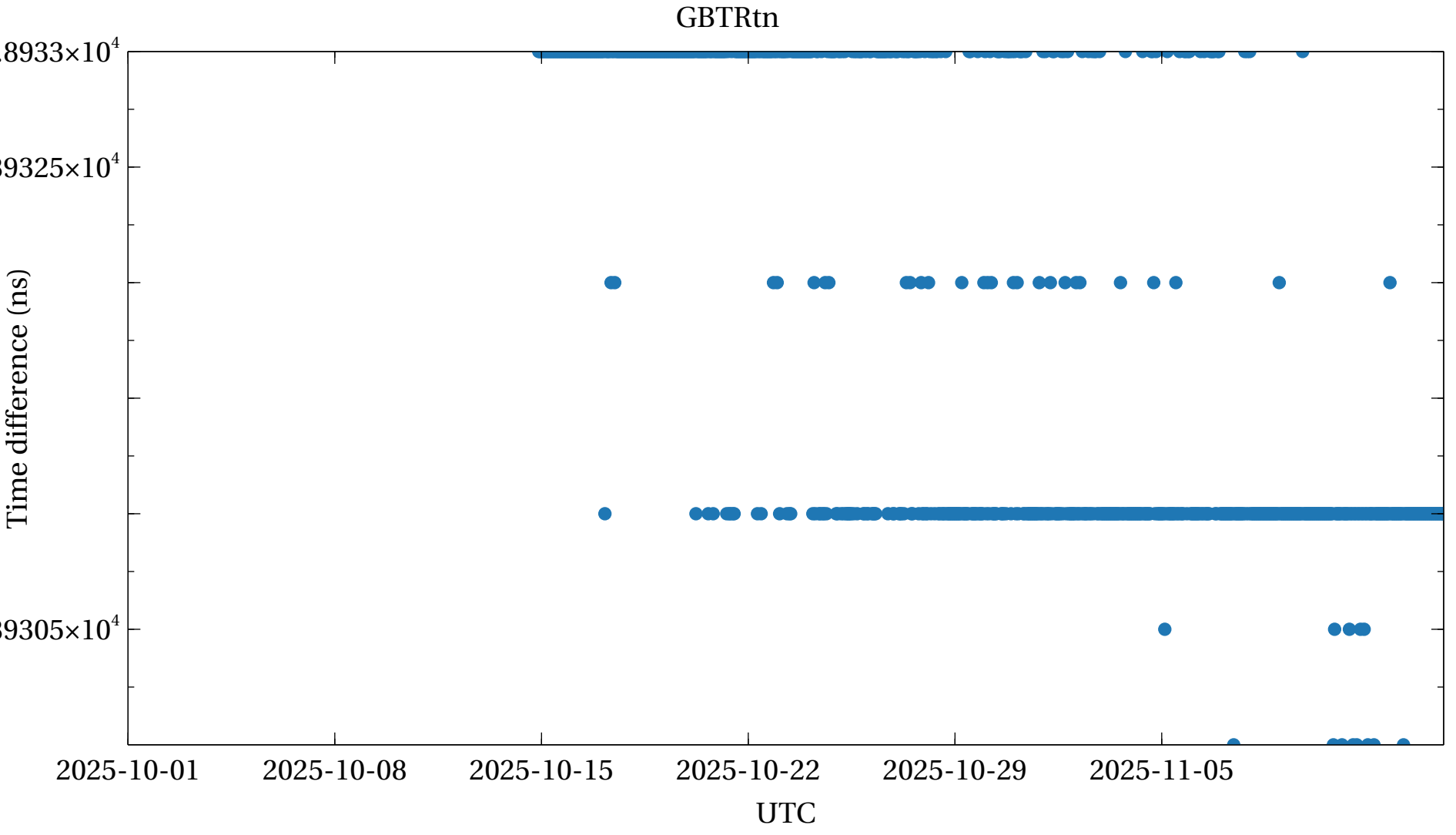

CNS2018; hourly medians
−2240
−2260
−2280
−2300
−2320
Time difference (ns)
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)

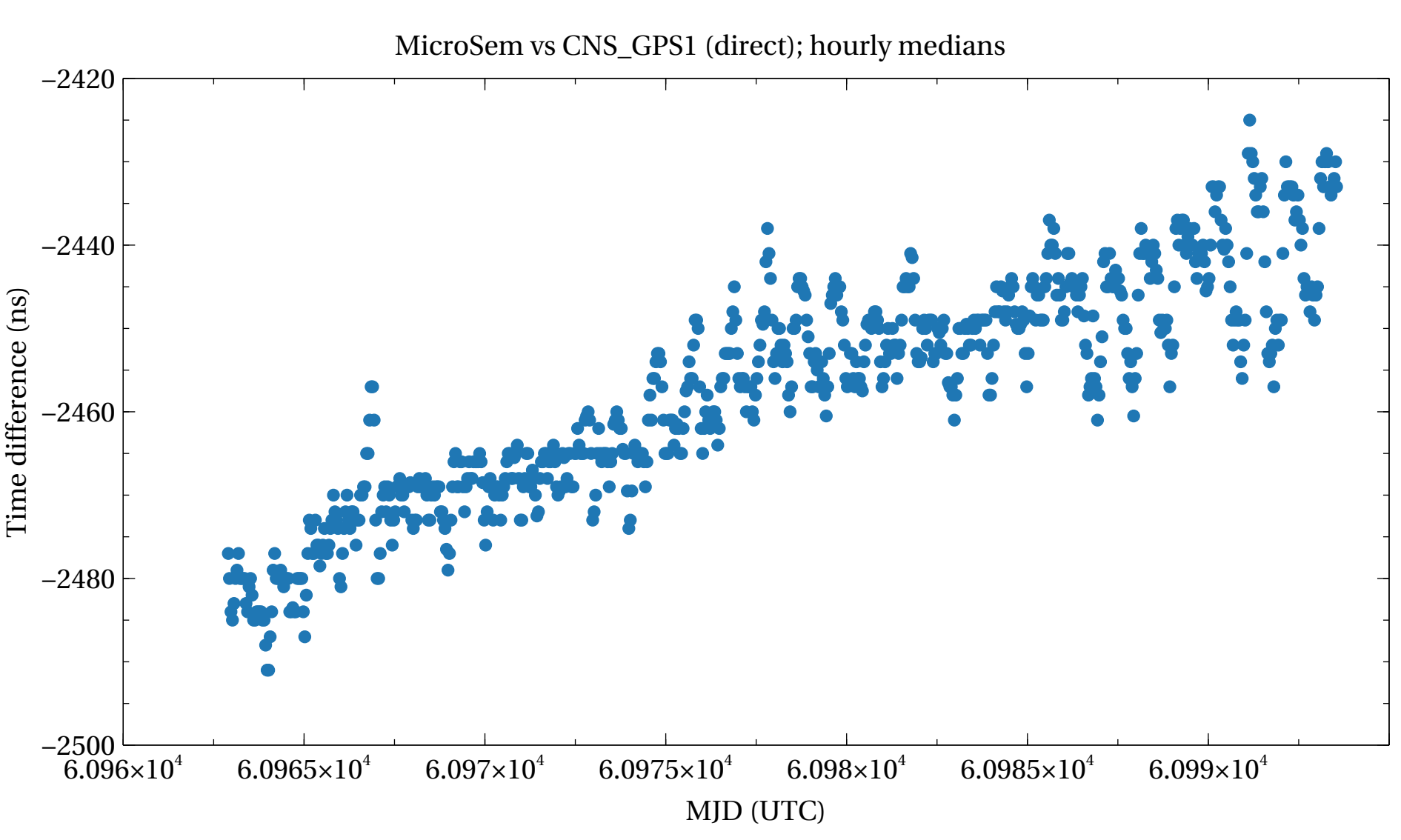

CNS2018
−2240
−2260
−2280
−2300
−2320
Time difference (ns)
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC

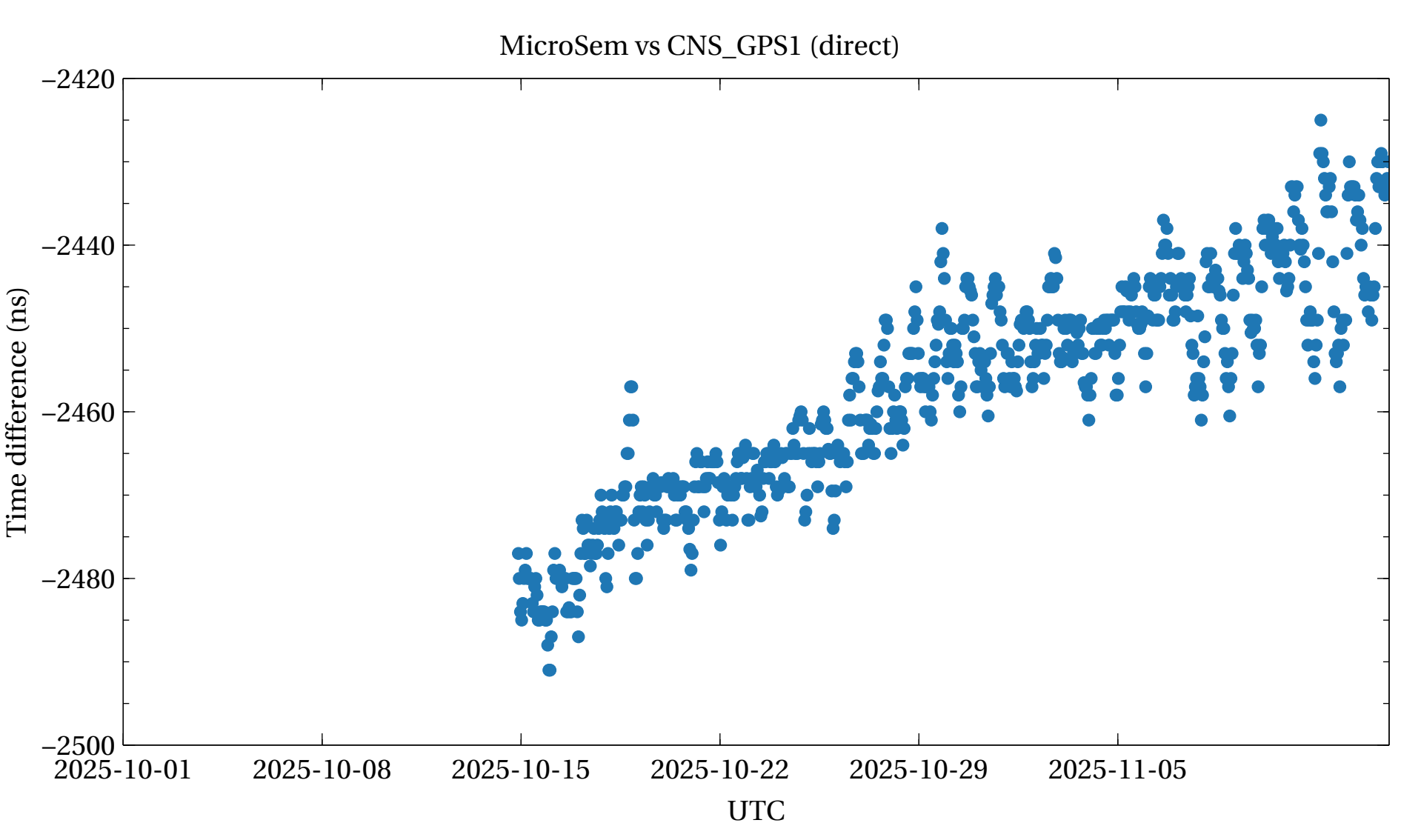

GBT_VLBA; hourly medians
Time difference (ns)
560
550
540
530
520
510
500
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)

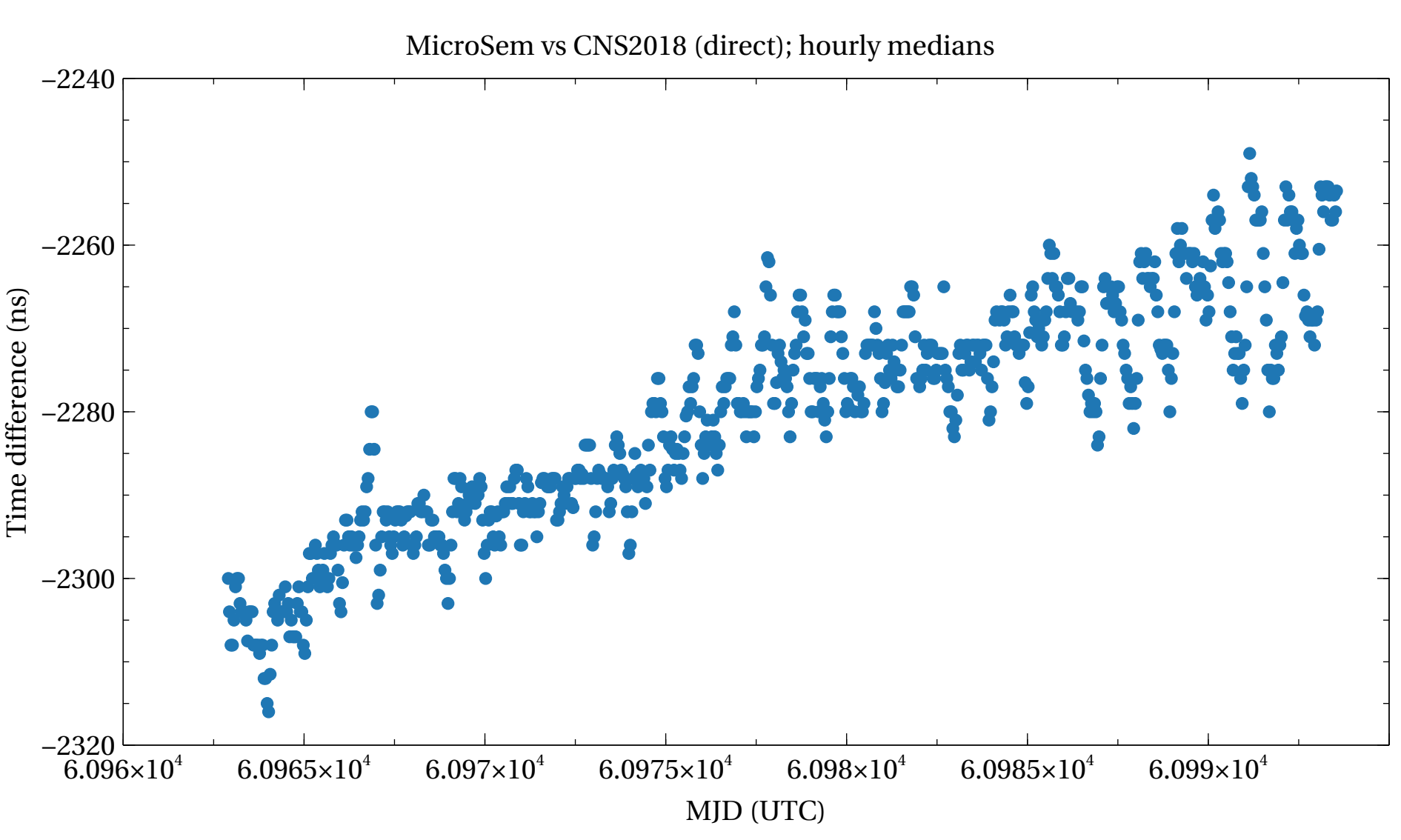

GBT_VLBA
560
550
540
530
520
510
500
Time difference (ns)
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC

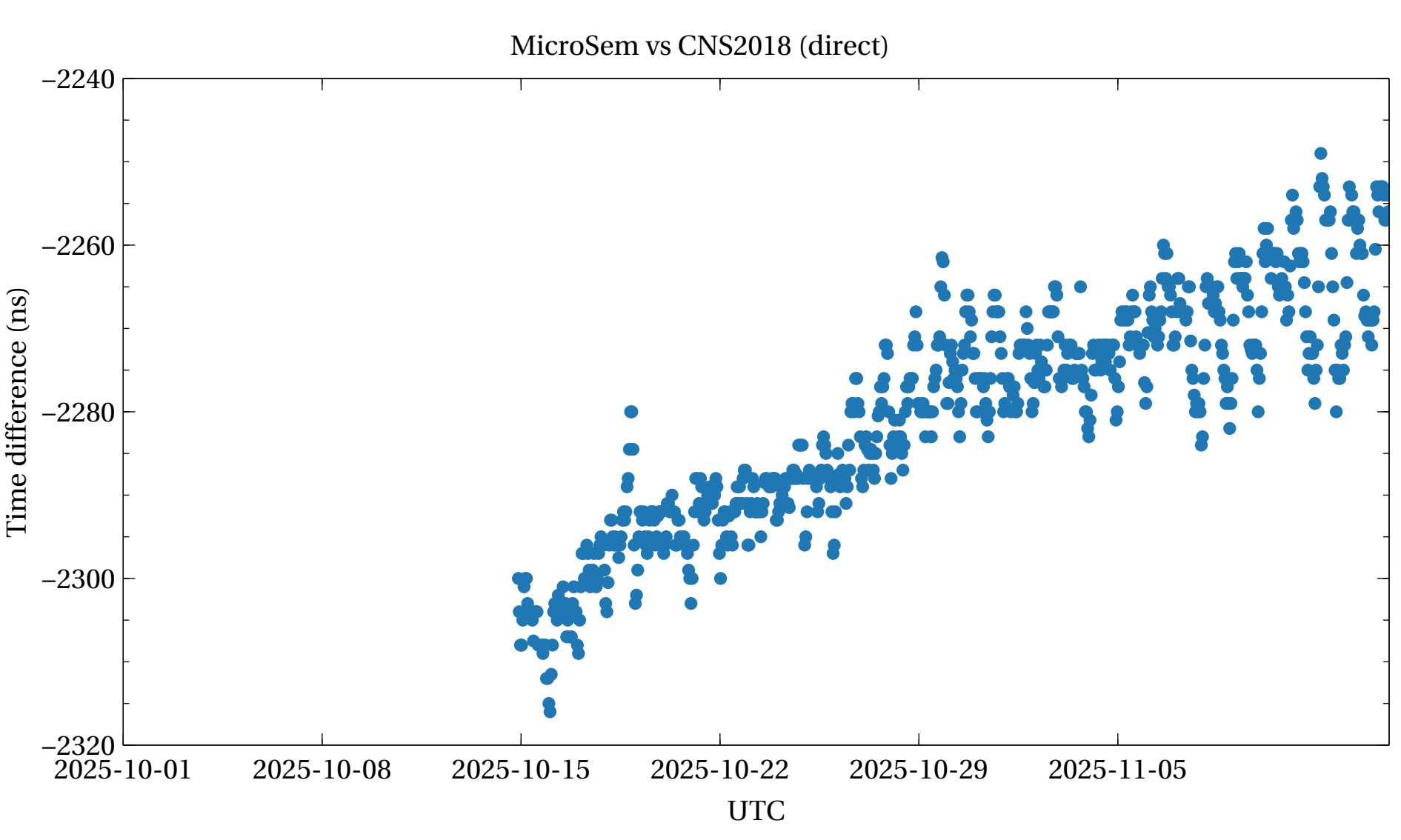

GBTRtn; hourly medians
Time difference (ns)
MJD (UTC)
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4

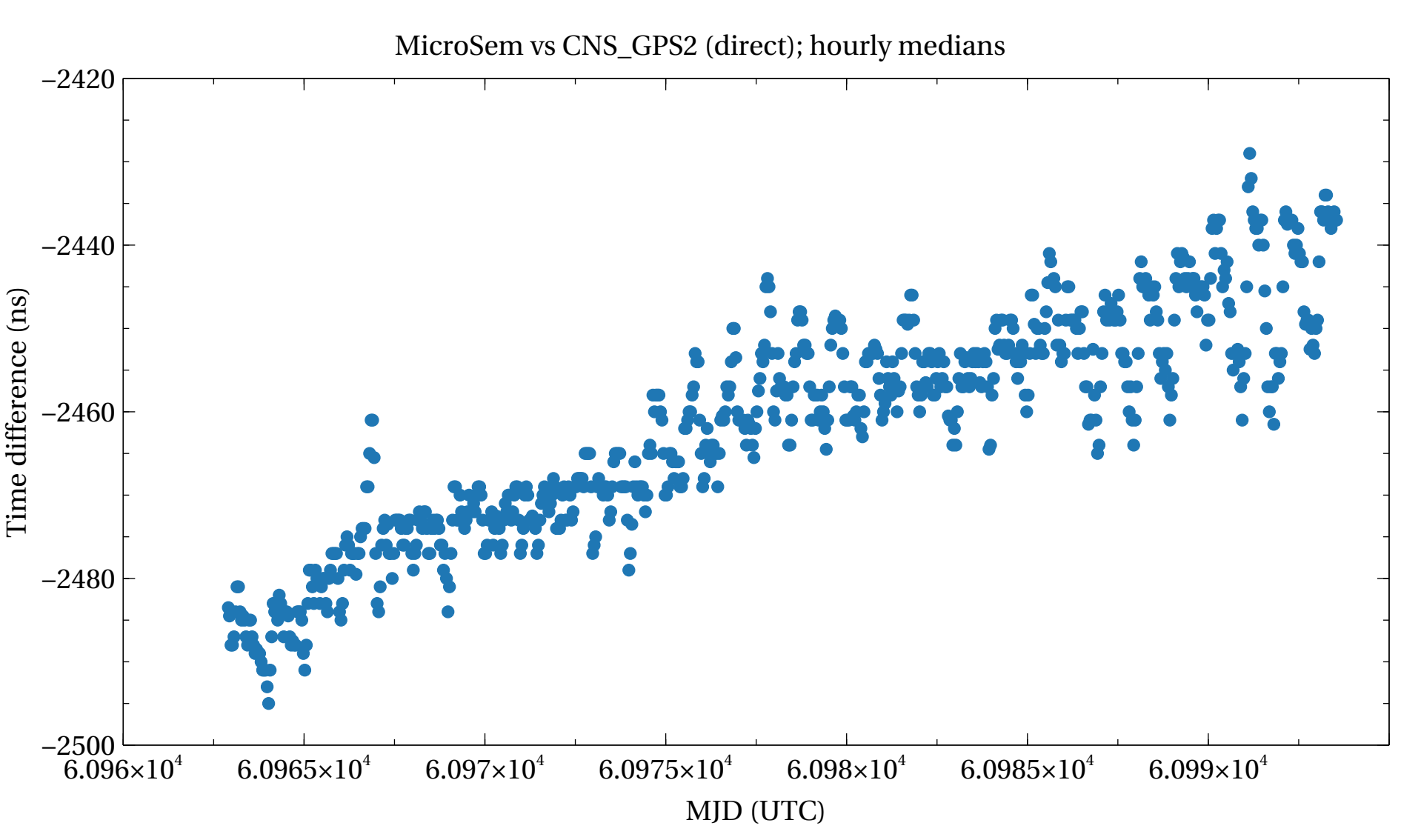

GBTRtn
8933×10^4
9325×10^4
9305×10^4
Time difference (ns)
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC

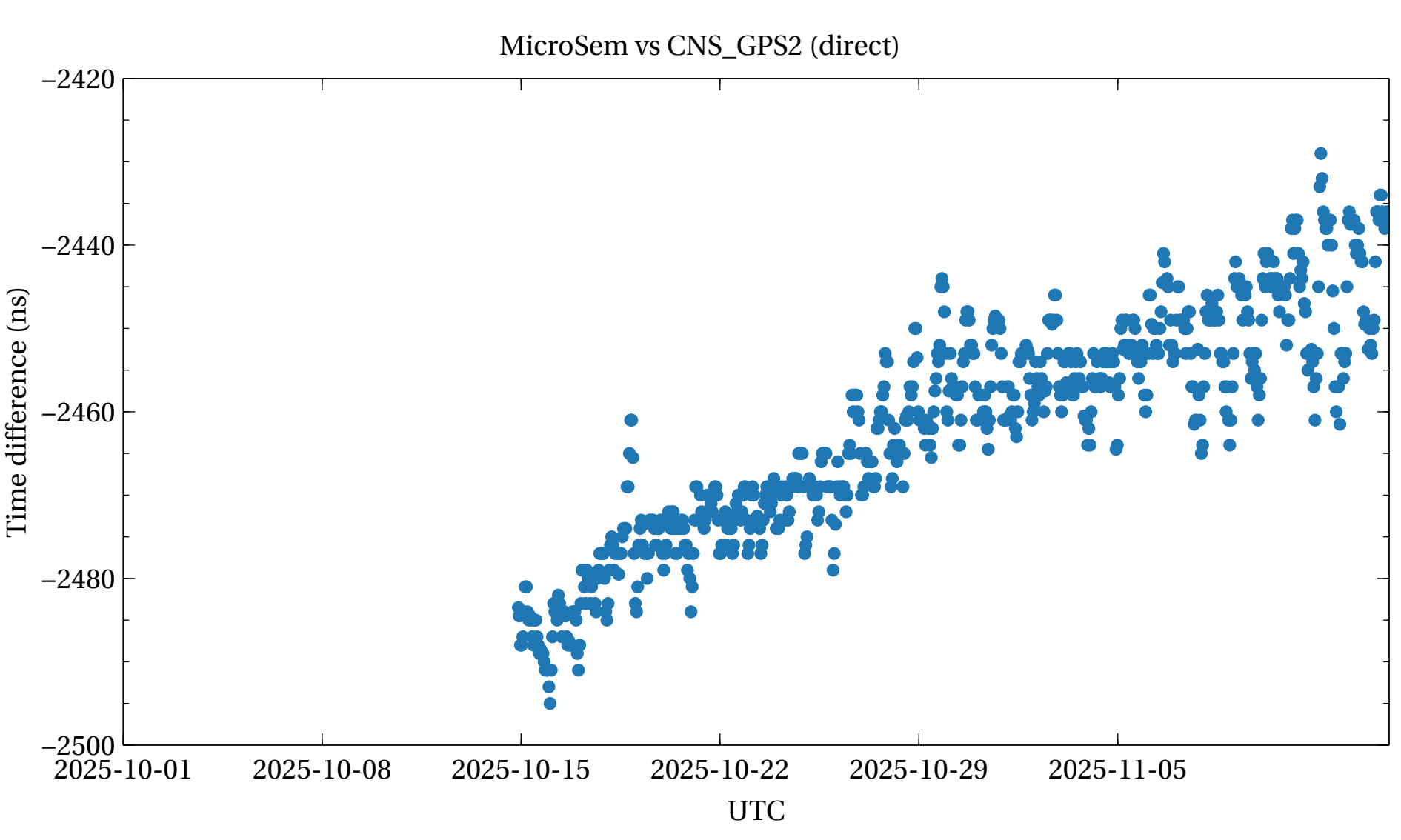

MicroSem vs CNS_GPS1 (direct); hourly medians
−2420
−2440
−2460
−2480
−2500
Time difference (ns)
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)

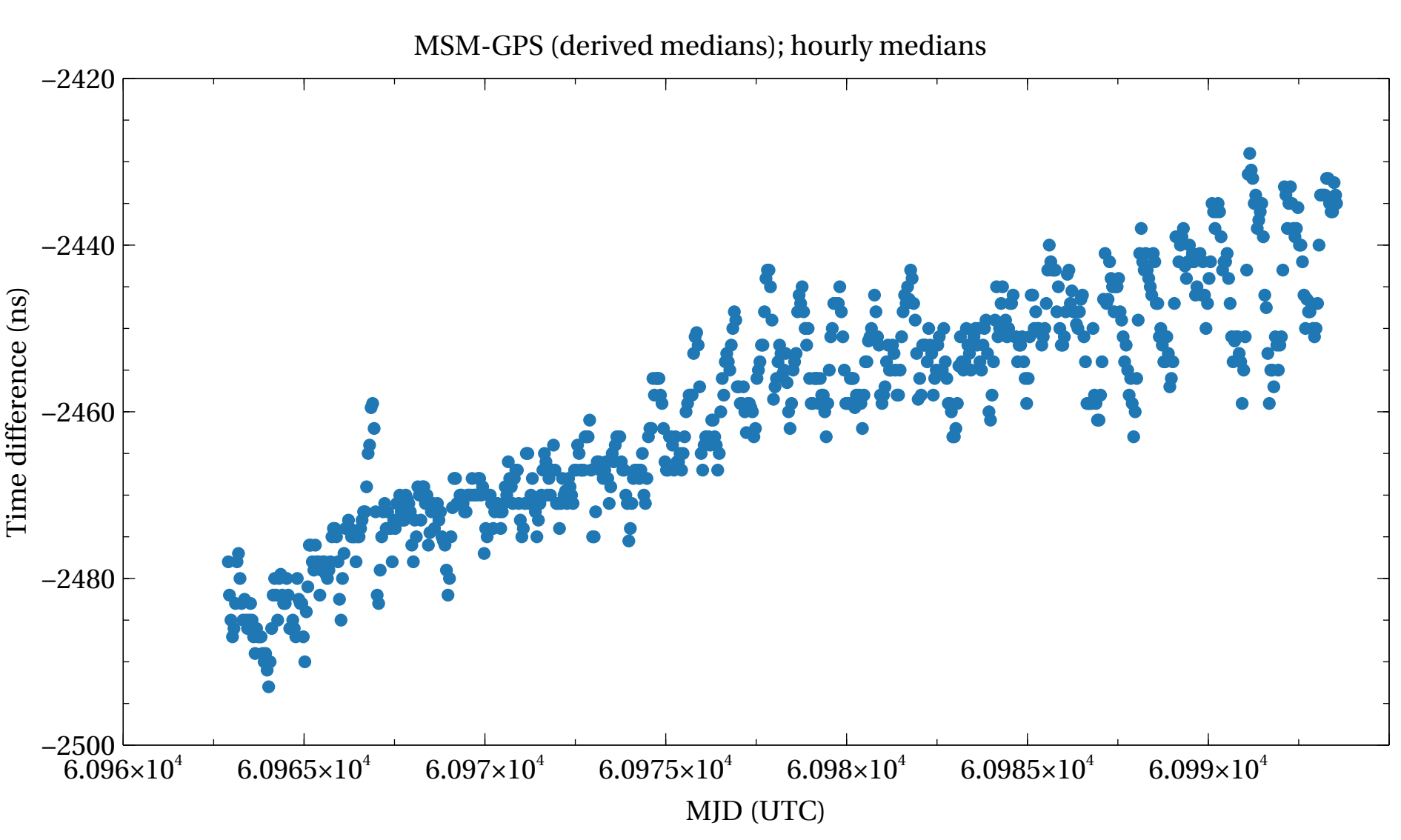

MicroSem vs CNS_GPS1 (direct)
−2420
−2440
−2460
−2480
−2500
Time difference (ns)
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC

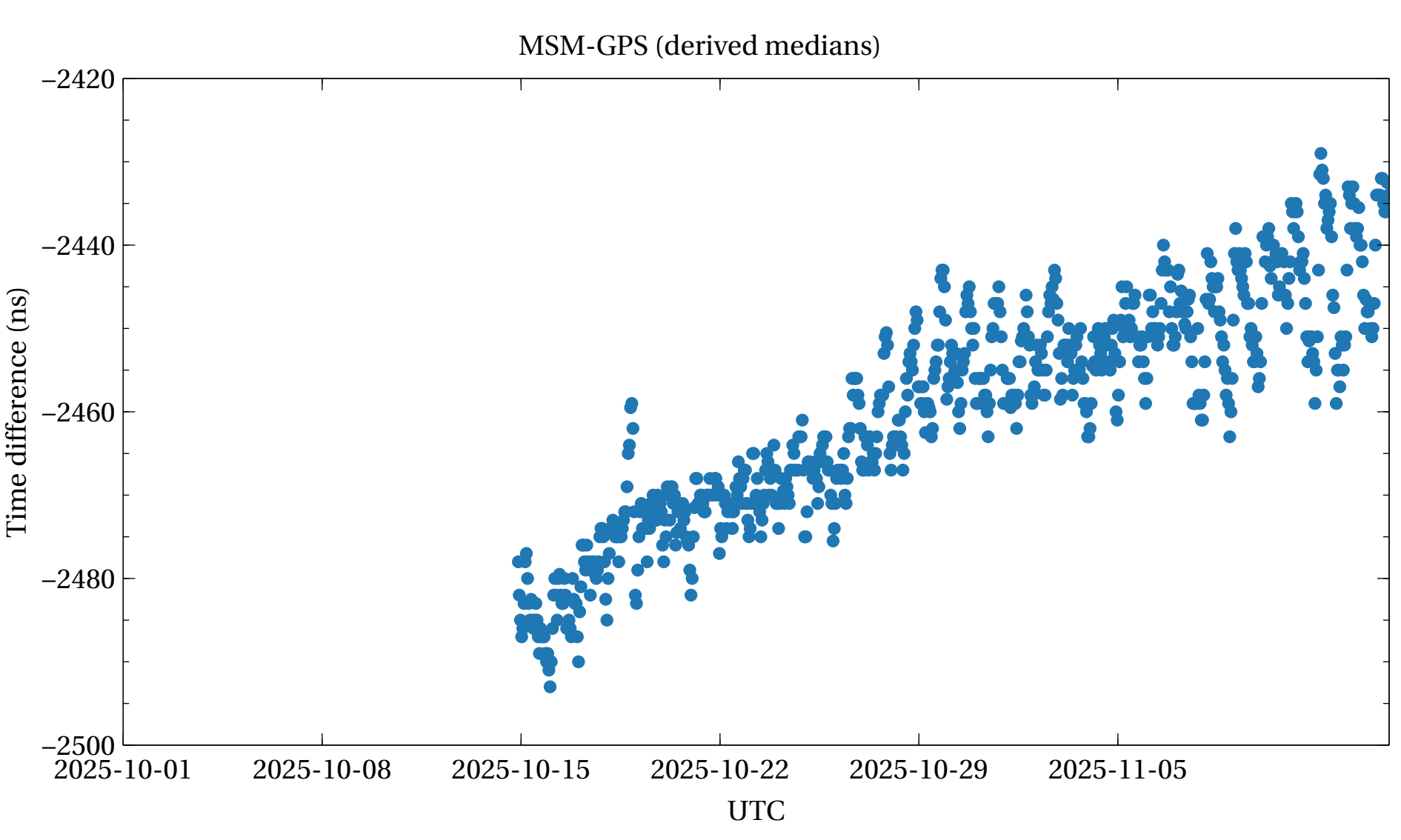

MicroSem vs CNS2018 (direct); hourly medians
−2240
−2260
−2280
−2300
−2320
Time difference (ns)
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)

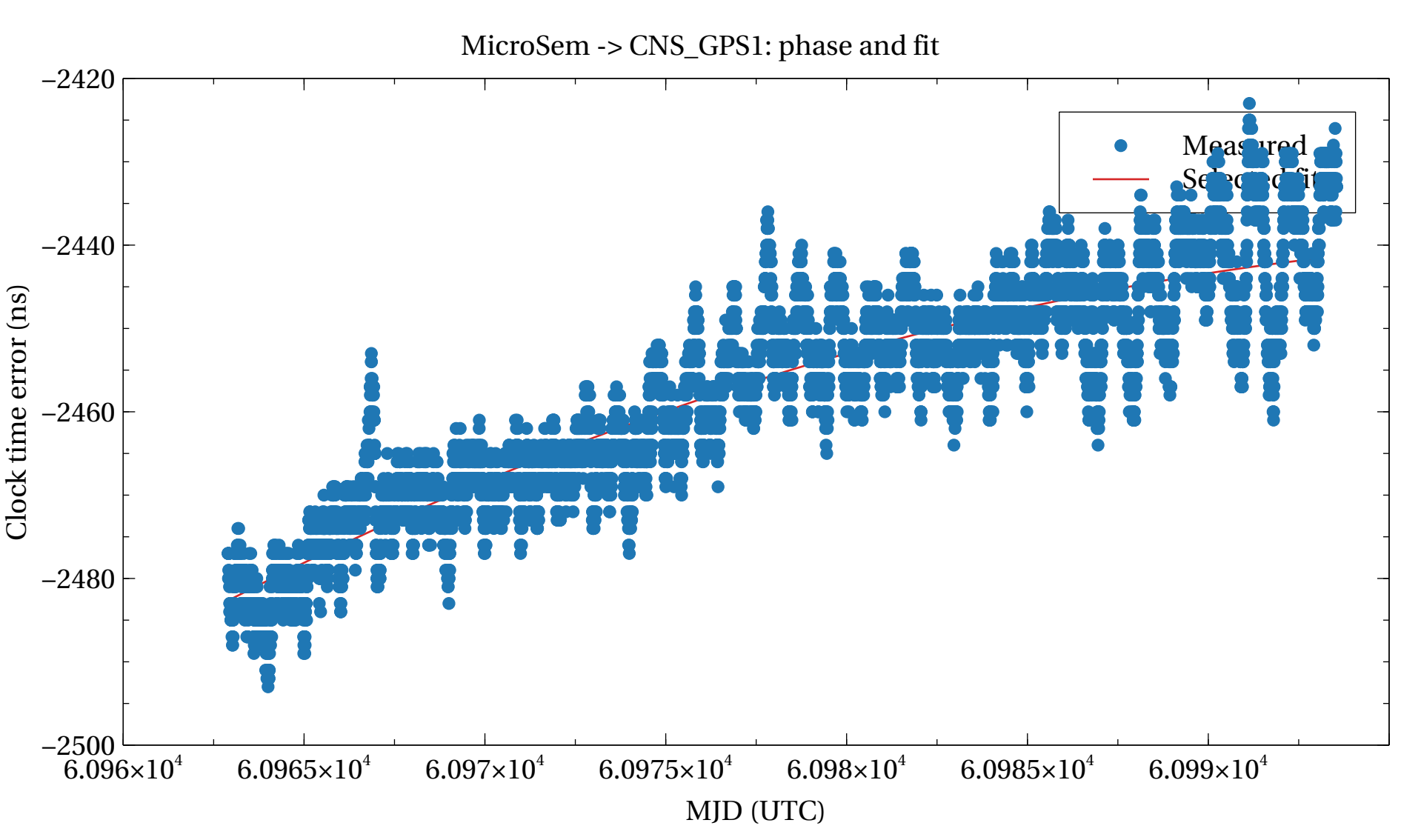

MicroSem vs CNS2018 (direct)
−2240
−2260
−2280
−2300
−2320
Time difference (ns)
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC

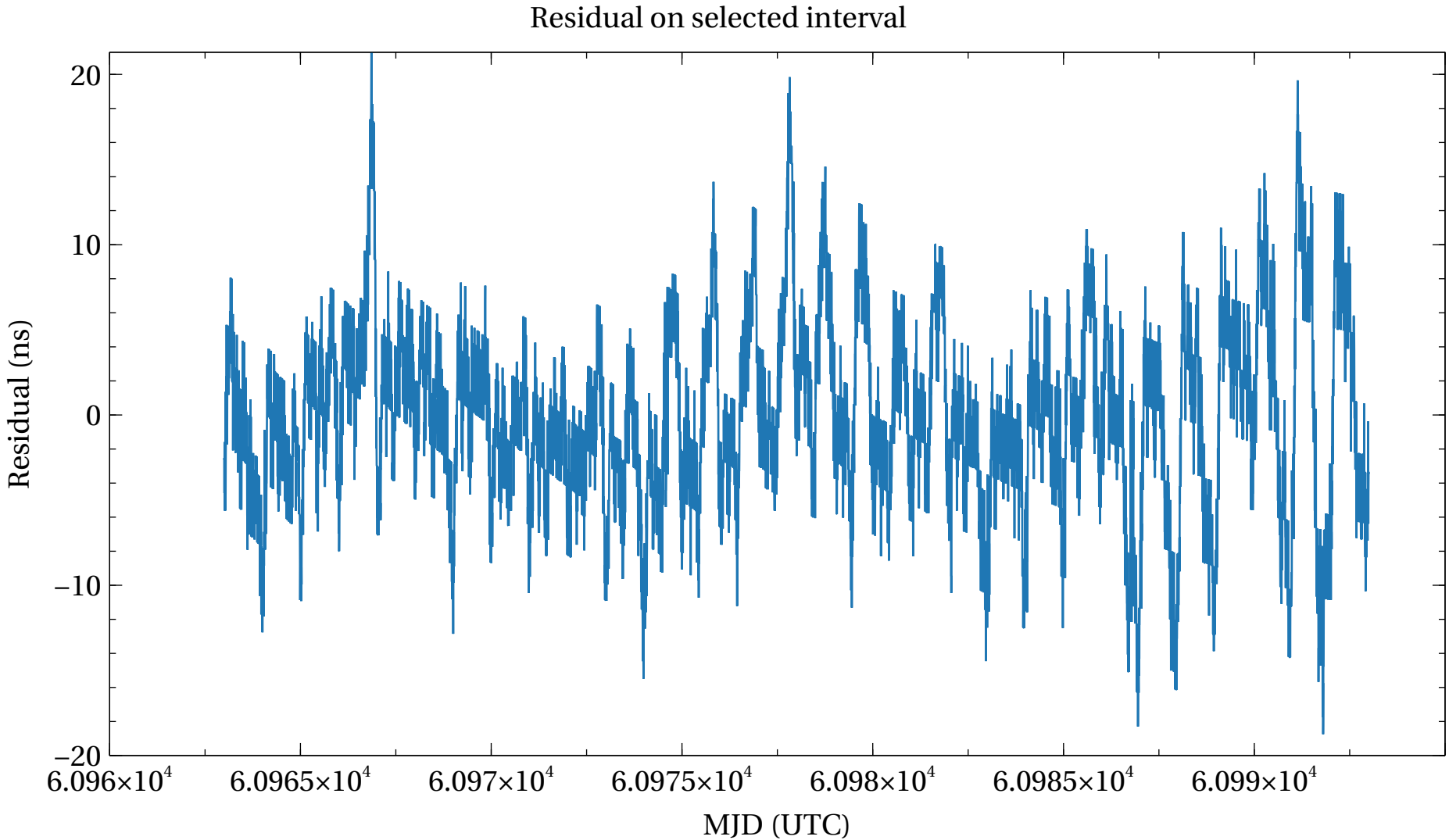

MicroSem vs CNS_GPS2 (direct); hourly medians
−2420
−2440
−2460
−2480
−2500
Time difference (ns)
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)

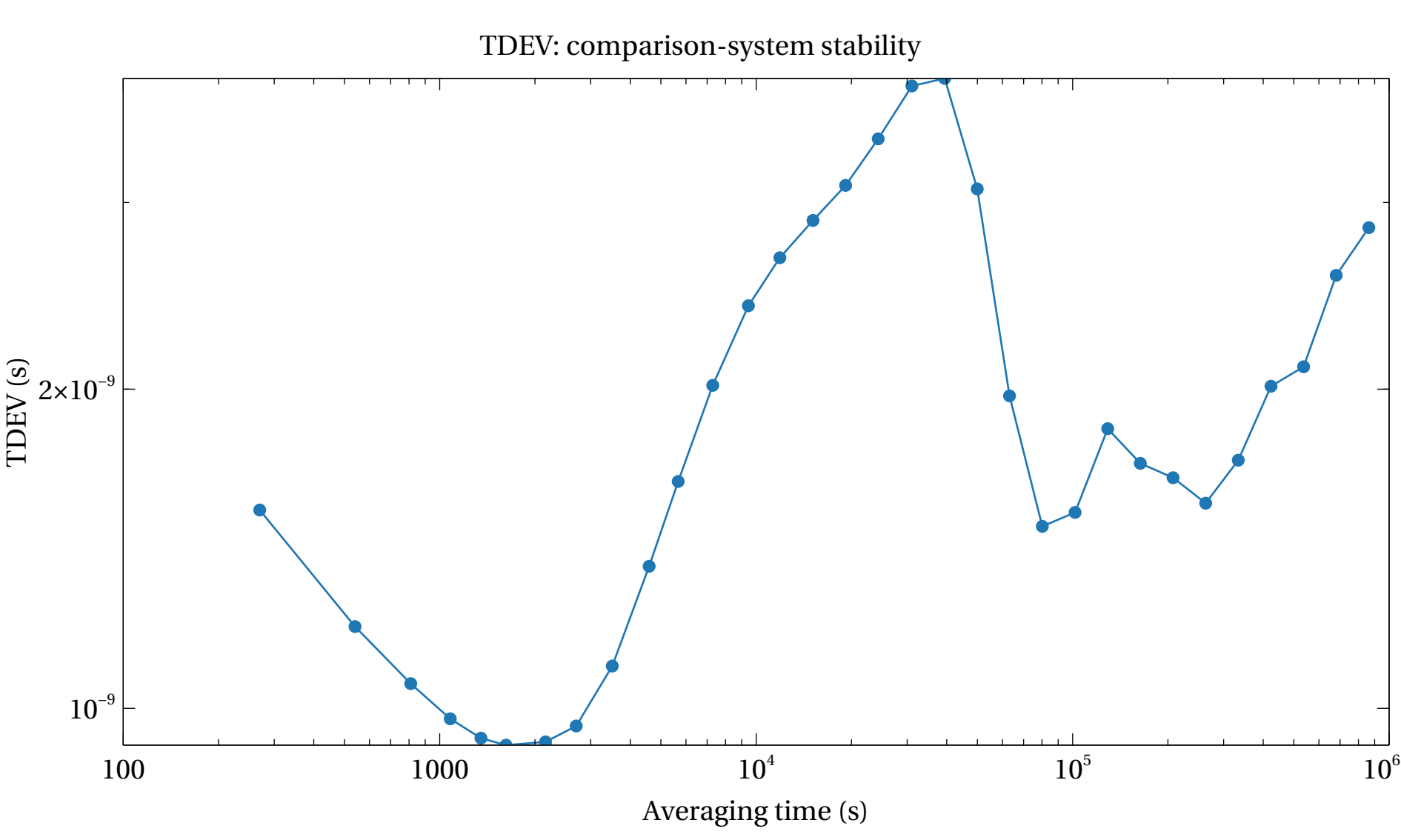

MicroSem vs CNS_GPS2 (direct)
−2420
−2440
−2460
−2480
−2500
Time difference (ns)
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC

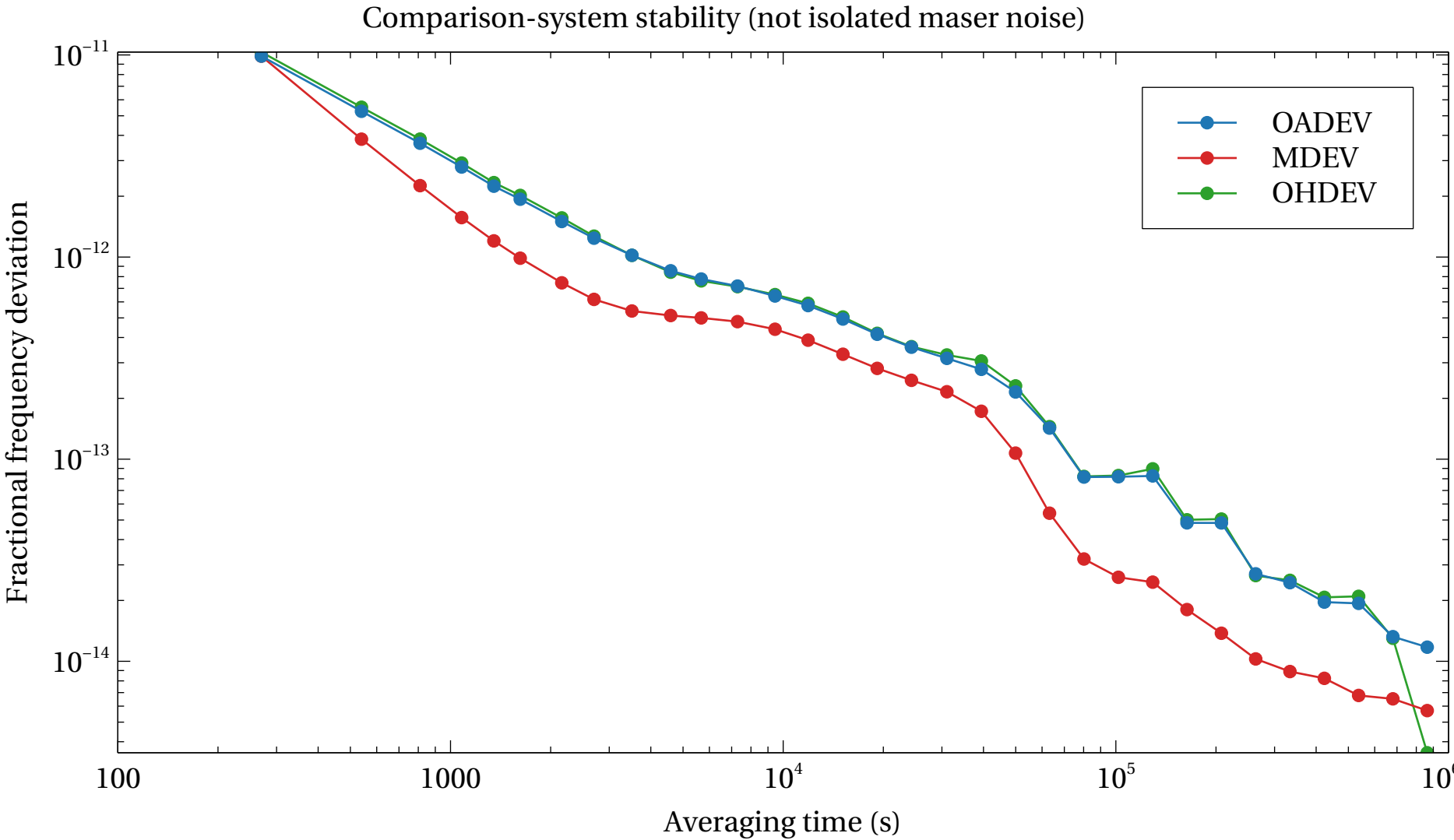

MSM-GPS (derived medians); hourly medians
−2420
−2440
−2460
−2480
−2500
Time difference (ns)
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)

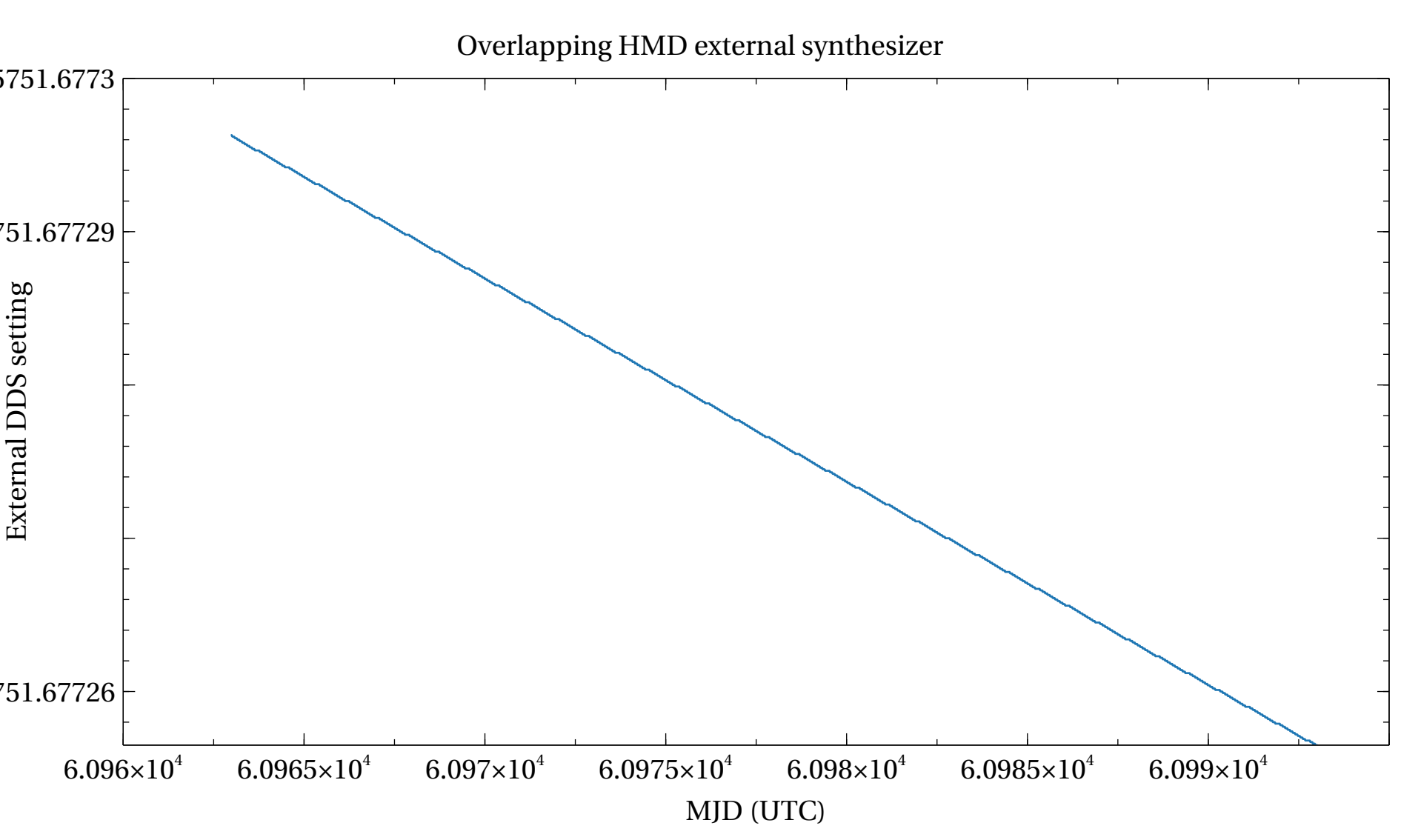

MSM-GPS (derived medians)
−2420
−2440
−2460
−2480
−2500
Time difference (ns)
2025-10-01
2025-10-08
2025-10-15
2025-10-22
2025-10-29
2025-11-05
UTC


MicroSem -> CNS_GPS1: phase and fit
Measured
Selected fit
−2420
−2440
−2460
−2480
−2500
Clock time error (ns)
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)


Residual on selected interval
Residual (ns)
20
10
0
−10
−20
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)


TDEV: comparison-system stability
TDEV (s)
2×10⁻⁹
10⁻⁹
100
1000
10⁴
10⁵
10⁶
Averaging time (s)


Comparison-system stability (not isolated maser noise)
Fractional frequency deviation
10^{-11}
10^{-12}
10^{-13}
10^{-14}
100
1000
10^4
10^5
10^6
Averaging time (s)
OADEV
MDEV
OHDEV


Overlapping HMD external synthesizer
5751.6773
751.67729
751.67726
External DDS setting
6.096×10^4
6.0965×10^4
6.097×10^4
6.0975×10^4
6.098×10^4
6.0985×10^4
6.099×10^4
MJD (UTC)